

Chapter 1

Explainability in Reinforcement Learning

Today, the majority of artificial intelligence agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences. Even more autonomous systems, such as agents based on large language models, remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both reinforcement learning and deep learning. As a result, the mechanisms underlying their decisions cannot be understood directly from the learned model. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, a research field has emerged: Explainable Reinforcement Learning (XRL), a subfield of Explainable Artificial Intelligence (XAI). The objective of XRL is to develop explainability methods that make it possible to obtain explanations of reinforcement learning agents.

This chapter presents the main existing XRL methods. We first define what is meant by an explanation in the context of XAI. We then propose a taxonomy for organizing existing XRL methods. This taxonomy serves as the basis for the state-of-the-art review presented in the remainder of the chapter.

1.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in a general.

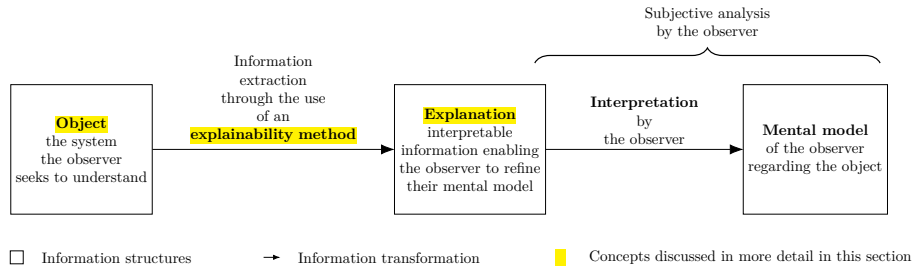


Figure 1.1: Explanation process in explainable artificial intelligence.

We then describe how this concept applies to XAI. Finally, we introduce the concept of explainability methods and describe the properties that characterize explanations in the context of XAI.

1.1.1 Explanation

Establishing what is meant by an explanation is challenging due to the fundamentally multidisciplinary nature of the concept. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI.

For this reason, we propose the following definition of explanation. An explanation involves two entities: the object (Definition 1.1.1) being explained and the receiver of the explanation. The receiver is a human who aims to understand how the object operates and will be referred to as the observer throughout this manuscript (Definition 1.1.2). The observer seeks to understand the object and, in doing so, builds a mental model of its functioning. An explanation is therefore what allows the observer to refine and correct their mental model of how the object operates (Definition 1.1.4).

For an observer to understand an explanation, the information it contains must be interpretable (Definition 1.1.3). Information is considered interpretable when it can be directly understood by a human without requiring any additional transformation process.

Definition 1.1.1: Object

The object of an explanation is the system is to be understood by an observer.

Definition 1.1.2: Observer

The observer is the human who aims to understand a object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

Definition 1.1.3: Interpretable

Information is interpretable when it can be directly understood by an observer without requiring any additional transformation process.

Definition 1.1.4: Explanation

An explanation is any interpretable information that enables an observer to refine or correct their mental model of the object of the explanation.

This definition of explanation provides the foundation for the approach to explainability adopted throughout this manuscript. Its generality allow it to encompass the different explainability methods considered in the following sections.

Moreover, it avoids a binary interpretation of explanation, where a system is considered either explainable or not explainable. From our perspective, the notion of non-explainable system is not meaningful. Considering a system as impossible to explain would therefore contradict the general objective of scientific understanding.

This quantitative perspective allows us to express an intuitive idea: not all explanations provide the same level of value. The effectiveness of an explanation depends on its ability to improve the observer's understanding of the system. For example, describing a marginal aspect of a system is different from correcting an observer's misunderstanding of a key mechanism.

Figure 1.1 summarizes this whole explanation process, from the object of the explanation to the mental model refined by the observer.

1.1.2 Explainable Artificial Intelligence

Now that the general framework of explanation has been established, we can examine how this concept applies in the context of artificial intelligence. In the context of XAI, the object of explanation is an artificial intelligence system. More precisely, XAI emerged as a response to the success of deep learning models. These models achieve high performance through connectionist architectures based on deep neural networks. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of neural networks cannot be directly associated with human-defined concepts. From this starting point, two types of approaches emerge:

- **Substitutive explainability:** In this approach, the underlying assumption is that there is nothing to explain within a neural network. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the so-called "black box" view of neural networks. Under this assumption, XAI consists of replacing as much of the deep learning system as possible with alternative AI methods.
- **Intrinsic explainability:** In this approach, it is argued that neural networks inherently contain mechanisms that can be explained. The

limitation is considered to lie not in the networks themselves, but in the absence of adequate tools to analyze their functioning.

In this manuscript, we present methods stemming from both approaches to explainability. This manuscript considers methods from both approaches in order to provide a comprehensive overview of the XAI field. Throughout this work, we adopt the intrinsic explainability perspective as the underlying viewpoint. This perspective influences how we define explainability and how we organize the analysis of existing methods.

1.1.3 Explainability Method

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. In our framework, an explanation in the field of XAI can be characterized by three components:

- **Source:** refers to the origin of the information used by the explainability method. The explainability method transforms the raw information produced by the AI system into an explanation that can be understood by an observer.
- **Form:** refers to the representation through which the explanation is communicated. The form defines how the information is presented to the observer and must allow human interpretation. An explanation can take different forms, such as text, images, computer code, mathematical equations.
- **Subject:** refers to the aspect of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

An explainability method can therefore be viewed as a transformation process that maps a source of information into an explanation characterized by a specific form and subject. Current research in XAI focuses on developing new methods capable of performing this transformation in order to overcome the lack of interpretability associated with deep learning models.

1.2 Taxonomy of XRL Methods

Now that the concept of XAI has been defined, we turn to its application in reinforcement learning, known as XRL. As in XAI, explanations in XRL are produced by explainability methods and can be described in terms of their subject, source, and form. This section reviews the possible variations of these three aspects in the specific context of reinforcement learning.

1.2.1 Subject

The first question to address is the subject of explanation in the context of reinforcement learning. To answer this question, it is necessary to identify the

different components involved in a reinforcement learning system and determine which of them constitutes the object of explanation. A reinforcement learning system can be described through two main components: the environment and the policy.

The environment defines the dynamics in which the agent evolves, including the possible states, observations, actions, and transitions. Environments are simulations designed by humans, and their dynamics are explicitly defined. Therefore, it is assumed that the environment is known to the observer and does not require explanation.

The main object of interest is therefore the agent’s policy. The policy represents the decision-making mechanism of the agent. It is a function that maps observations to actions and is learned through interaction with the environment in order to maximize the expected cumulative reward. The policy is the component that encodes the learning process and that XRL aims to explain.

Explaining a policy introduces additional challenges compared with classical XAI settings. In traditional XAI problems, the objective is to explain an isolated decision, such as determining which features of an image led to a classification result. In contrast, reinforcement learning introduces a temporal dimension into the decision-making process. An action is not only determined by the current observation but also influences future states and rewards. Therefore, understanding the behavior of an agent may require considering a sequence of interactions between the agent and its environment.

This temporal dimension leads to a distinction between two objectives of explanation in XRL:

- **Decision explanations:** focus on identifying the elements of the current observation that influenced the action selected by the policy. This type of explanation is close to classical XAI approaches, as it analyzes the relationship between inputs and outputs while abstracting away the sequential nature of the decision process.
- **Strategy explanations:** aim to characterize the agent’s behavior over time. They focus on sequences of decisions and interactions with the environment in order to understand how the agent achieves its long-term objective. This type of explanation is specific to reinforcement learning, as it addresses the temporal and goal-oriented nature of the learning process.

1.2.2 Source

We identify three sources of explanation in reinforcement learning, all obtained during or after the learning process and containing information about the agent:

- **Policies:** policies are functions that generate a distribution over the action space for a given observation. The objective of reinforcement learning is to learn a policy that maximizes the expected cumulative future rewards. During the learning phase, the policy is iteratively improved through interactions with the environment.
- **Trajectories:** trajectories represent sequences of successive interactions between the agent and the environment. At each step, the agent selects an action based on its observation, which modifies the environment and

180 produces a reward as well as a new observation. This process continues
181 until the end of the trajectory.

- 182 • **Value functions:** value functions provide predictions about the future
183 outcome of a trajectory from a given observation. The most commonly
184 used value function is the critic function, which estimates the expected
185 sum of future rewards. This prediction guides the learning process, as the
186 agent updates its policy to maximize the estimated future return. However,
187 other types of value functions can also be used as sources of explainability.

188 1.2.3 Form

189 We identify six forms of explanation in the RL setting:

- 190 • **Policy Model:** an interpretable model of the agent’s policy function.
- 191 • **Saliency Map:** a weighting of the input features representing their
192 influence on the output of the policy function.
- 193 • **Counterfactual:** a representation of a modified observation obtained by
194 perturbing the original observation in order to produce a different action
195 from the agent.
- 196 • **Agent Trajectory:** a representation of a sequence of successive interac-
197 tions between the agent and the environment.
- 198 • **Trajectory Prediction:** a prediction about the future evolution of the
199 trajectory produced by a value function.
- 200 • **Interaction Model:** an interpretable model of the interactions between
201 the agent and the environment.

202 Among the three components characterizing an explanation, the form is the
203 one that varies most significantly across explainability methods. While subjects
204 and sources tend to be relatively stable in the XRL literature, the diversity in
205 how explanations are presented is considerably greater. The list above is by no
206 means definitive and may evolve as new approaches emerge. For this reason, the
207 following section is organized according to the form of explanations and presents
208 existing explainability methods based on this criterion.